

Aeson Tran

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EDUCATION

University of California: San Diego

La Jolla, California

B.S., Computer Engineering

Expected Graduation, June 2029

- o **GPA:** 3.925
- o **Related Coursework:** Data Structures & Algorithms, Systems Programming & Software Tools, Linear Algebra, Differential Equations, Computer Organization

Mission Hills High School

San Marcos, California

High School Diploma

Aug 2021 – June 2025

- o **GPA:** 4.82

Palomar College

San Marcos, California

Student

June 2024 – June 2025

- o **Related Coursework:** Object-Oriented Programming, Multivariable Calculus, C Programming, C++ Programming, Java Programming

RESEARCH AND PROJECTS

Planetary Physics Engine

Dec 2024 – June 2025

- Utilized C++ and OpenGL to simulate real-life representations of multi-body planetary orbit and black holes
- Acquired skills in modular programming, tensor math implementations, and code optimization
- Integrated 3D graphical rendering and window management to enhance user-friendliness and ease-of-use

Server Emulator

June 2023 – Jan 2024

- Assisted in creating and finalizing video game proxy emulation software compatible with both mobile and desktop environments with 240+ users
- Collaborated with 13 others to advance skills in data serialization with Google Protobufs, full-stack development, and system integration between teams
- Wrote multiple emulators in a variety of languages, ranging from Java to Zig and integrated programs such as MongoDB Community Server and a proprietary SDK server written in Rust

All-Purpose Discord Bot Programs

Dec 2022 – Apr 2023

- Developed bot-like applications through Discord.py and Discord.js libraries for 10+ Discord servers
- Implemented features such as music playback and user lookup requiring efficient and safe database integration with MongoDB and API connections such as Spotify, Discord, and FFLogs to serve 100+ users per server
- Provided missing Discord features through personalized and interactive commands such as board games and currency

ACTIVITIES AND LEADERSHIP

Cardinal Alliance Marching Band

San Marcos, California

Drum Major

July 2024 – Dec 2024

- Managed and collaborated with a team of band board members and section leaders, delegating responsibilities to streamline rehearsal workflow
- Coordinated communication between directors/staff and student body to gather feedback and improve productivity
- Mentored new members and maintained high morale while ensuring focus, helping reach a school record of 90.10

I AM AP Club

San Marcos, California

Mathematics Coordinator, Ambassador

Aug 2022– June 2025

- Mentored 40+ students, dedicating 18+ hours per semester to academic guidance
- Coordinated study events and collaborative breakout sessions with board members to provide missing learning opportunities
- Assisted in increasing average math test scores by 5% in 2025 through personalized study plans for every student

SKILLS

Programming: C++, C, Java, React.js, Rust, Python, HTML/CSS, Lua, Node.js, MATLAB

Tools: IntelliJ, PyCharm, CLion, CMake/Make, Git, MongoDB, Visual Studio, Visual Studio Code, Vim/Neovim